



# Boron nitride doped $\text{Li}_7\text{P}_3\text{S}_{11}$ solid electrolyte with improved interfacial compatibility and application in all-solid-state Li/S battery

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## Abstract

All-solid-state lithium sulfur battery is considered as one of the most promising candidates for the future application due to high theoretical energy density and enhanced safety. However, the all-solid-state lithium sulfur battery based on  $\text{Li}_7\text{P}_3\text{S}_{11}$  solid electrolyte has serious interface issues between electrode and solid electrolyte. The reaction of lithium metal and  $\text{Li}_7\text{P}_3\text{S}_{11}$  solid electrolyte would form a layer without lithium-ion conducting, which could impend the lithium-ion transfer and transport during cycling and causes increase of internal resistance and capacity fading. Herein, we introduce a solid electrolyte  $\text{Li}_7\text{P}_3\text{S}_{11}$  with BN doping for the all-solid-state lithium sulfur battery. The BN nanoflakes could partially isolate the  $\text{Li}_7\text{P}_3\text{S}_{11}$  solid electrolyte and lithium metal, thus improving the interfacial compatibility of electrode and electrolyte. With BN doped  $\text{Li}_7\text{P}_3\text{S}_{11}$  used as solid electrolyte, the all-solid-state Li/S battery exhibits an enhanced electrochemical performance with better cycling stability and higher Coulombic efficiency.

## 1 Introduction

With the increasing demand for electric vehicles and grid energy storage, high energy density and improved safety of batteries are required [1–3]. The lithium/sulfur (Li/S) battery has been captured much attention due to high theoretical energy density (2600 Wh/kg), high abundance and low cost of sulfur [4]. However, the Li/S battery with liquid electrolyte suffer the problems of short cycle life, low charging efficiency, poor safety and a high self-discharge rate, which are related to the “shuttle effect” of the lithium polysulfide and use of organic electrolyte [5–7]. To solve these issues, the all-solid-state Li/S battery has been proposed. With use of the solid electrolyte, it could successfully eliminate the

“shuttle effect”, inhibit the lithium dendrite growth and solve safety problems from organic electrolyte [8–10]. Therefore, the all-solid-state Li/S battery is considered as one of the most promising candidates for the further applications.

The solid electrolyte is the key for greatly enhancing the electrochemical performance of the all-solid-state Li/S battery.  $\text{Li}_7\text{P}_3\text{S}_{11}$  (LPS) solid electrolyte has attracted much interest due to its high ionic conductivity, wide electrochemical window and facile synthesis [11–13]. However, it would react with lithium metal because of its high chemical/electrochemical activity, forming a Li-ion insulation layer. The insulation layer results in the increase of the internal resistance of the cell, causing the capacity fading [14, 15]. Various methods have been carried out to improve the interface compatibility of LPS and metallic lithium. For example, the Li-In alloy is used as substitute for lithium metal in all-solid-state Li/S battery [16]. Besides, a LiF (LiI) layer is deposited on the LPS solid electrolyte and isolates the contact between LPS and Li metal, thus improving the interfacial compatibility of solid electrolyte and electrode [17].

BN, similar to graphene, is a layered structure with great electronic insulation performance and chemical/electrochemical stability [18, 19]. Herein, we introduce a BN doped LPS (LPS-BN) solid electrolyte for the all-solid-state Li/S battery. With the layered structure of BN, it could successfully partially isolate the direct contact of LPS and Li metal, thus suppressing the reaction and improving the interface

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stability. Therefore, the all-solid-state Li/S battery with LPS-BN solid electrolyte exhibits an enhanced electrochemical performance.

## 2 Experimental

### 2.1 Materials synthesis

The  $\text{Li}_2\text{S}$  (99.9%, Alfa Aesar) and  $\text{P}_2\text{S}_5$  (99.9%, Aladdin) (7:3 mol ratio) was mixed with BN nanoflakes (99%, XFNANO) in a different mass ratio (1%, 2% and 3%), and then it was added in acetonitrile solvent. After stirring for 72 h at 50 °C, it was dried at 80 °C for overnight. Then the obtained powder was annealed at 250 °C for 1 h and the BN doped LPS solid electrolytes were obtained. All preparation processes were performed in an argon atmosphere with water and oxygen content below 0.1 ppm.  $\text{Li}_7\text{P}_3\text{S}_{11}$  solid electrolyte was synthesized with the method mentioned above without BN doping. It was also further used as the electrolyte in cathode composites.

The cathode composite was made from mixing sulfur (99.99%, Aladdin), carbon black (99%, Sigma-Aldrich) and LPS solid electrolyte. Prior to the mixing, the sulfur powder and graphene (99%, XFNANO) were mixed with mass ratio of 1:1 and ball milling for 1 h at room temperature. Then the powders were heat-treated at 155 °C in air and kept for 10 h to get S/G composites. The S/G powder, LPS and super P was mixed with 50:40:10 mass ratio. After the mixture was ball milling at 580 rpm for 24 h in Ar ambient at room temperature, the cathode composites were obtained.

### 2.2 Material characterization

X-ray diffraction (XRD) measurements were carried out by a Miniflex 600 instrument with copper  $\text{K}\alpha$  radiation between 10° and 50°. The morphologies of the samples were characterized using scanning electron microscope (SEM) (FE-SEM Hitachi SU-70) and a transmission electron microscope (JEOL JEM-2100F). The X-ray photoelectron spectroscopy

(XPS, ESCALAB 250) analysis was carried out with Al  $\text{K}\alpha$  as the X-ray source.

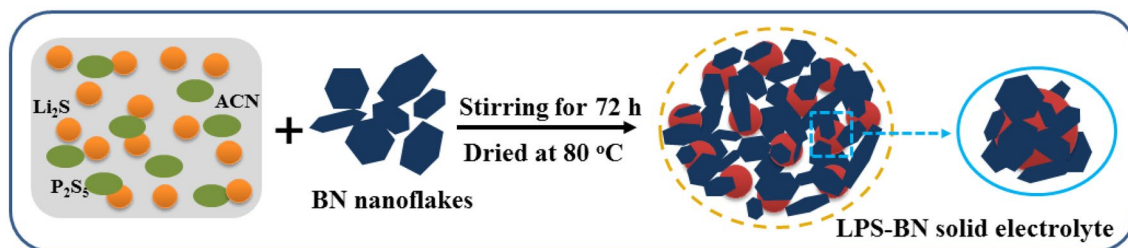
### 2.3 Electrode fabrication and electrochemical measurements

The ionic conductivities of synthesized LPS and BN doped LPS electrolytes were measured using electrochemical impedance spectroscopy (EIS) by a blocking symmetric with indium (In) foil as the electrode. The solid electrolyte powder was cold-pressed under 360 MPa with a thickness of ~1 mm in diameter of 10 mm. In foil was used as the electrode on both sides of the solid electrolyte pellets. All-solid-state Li/S batteries were fabricated by employing cathode composite and LPS solid electrolyte in a 10 mm die and then cold pressed at a pressure of 300 MPa. Then the lithium foil was attached on the other side of the electrolyte pellet and pressed at 120 MPa. All the preparation processes were performed in a dry argon-filled glovebox with  $\text{H}_2\text{O}$ ,  $\text{O}_2 < 0.1$  ppm.

The EIS measurements were carried out in the frequencies range from  $10^6$  Hz to 1 Hz from 25 to 80 °C by an electrochemistry workstation (Autolab analyzer PGSTAT302 N). Cyclic voltammogram was measured on the Chenhua analyzer (chi660e) at a scan rate of 0.1 mV/s between 1 and 3 V. The galvanostatic discharge–charge cycles were performed using a battery testing system (Land battery testing instrument) in a voltage of 1–3 V at room temperature.

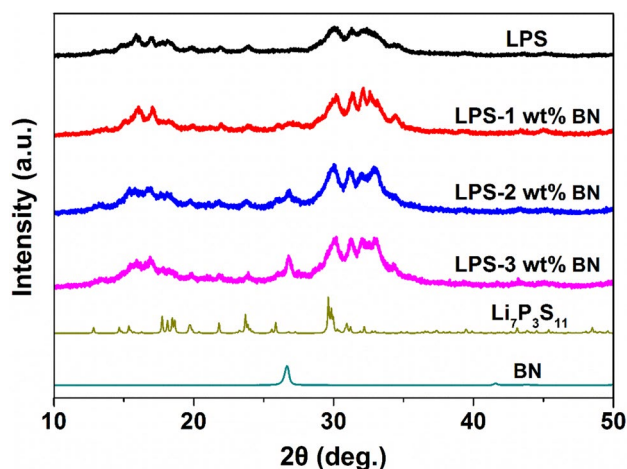
## 3 Results and discussion

The BN doped LPS solid electrolyte with different contents is synthesized by liquid-phase reaction method, as shown in Fig. 1. After stirring in ACN solvent for 72 h and dried at 80 °C for overnight, the BN nanoflakes coated LPS structured solid electrolyte is synthesized. As illustrated in Fig. 1, the BN nanoflakes would be coated on the LPS particles surface, isolating the direct contact between LPS and lithium metal in Li/S batteries. As the interface reaction could be suppressed by BN coating, the electrochemical performance



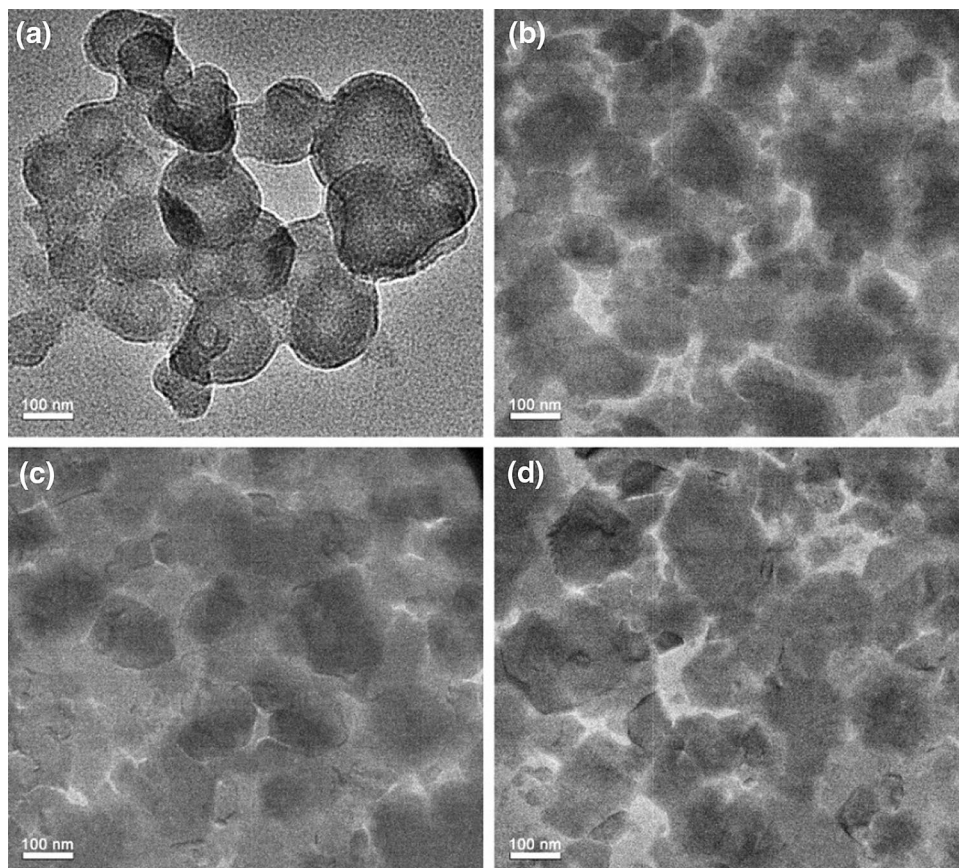
**Fig. 1** a Schematic of fabrication process of LPS-BN solid electrolyte

is successfully enhanced. The interfacial compatibility of LPS-*x* wt% BN and lithium metal gets more stable with increase of BN content. However, the ionic conductivity of the LPS-*x* wt% BN solid electrolyte decreases with more BN doping, which has a negative effect on the cell performance. Therefore, the BN content in LPS-*x* wt% BN solid



**Fig. 2** XRD spectra of LPS and LPS-*x* wt% BN solid electrolytes with *x* varies from 1 to 3%

**Fig. 3** TEM image of LPS and LPS-*x* wt% BN solid electrolytes with *x* is **a** 0%, **b** 1%, **c** 2%, **d** 3%



electrolyte has an effect on the electrochemical performance of the all-solid-state Li/S battery, which will be discussed in more details in the following sections.

The XRD measurement of LPS and LPS-BN solid electrolytes has been taken out and the patterns are shown in Fig. 2. From the XRD patterns, the main diffraction peaks at  $2\theta = 15.9^\circ, 29.8^\circ, 30.0^\circ, 31.2^\circ, 31.9^\circ, 34.3^\circ$  can be indexed to  $\text{Li}_7\text{P}_3\text{S}_{11}$ . And the diffraction peaks of  $2\theta = 26.8^\circ$  and  $41.6^\circ$  are ascribed to BN (JCPDS card No. 34-0421). It clearly displays that the BN could not react with LPS solid electrolyte during the preparation process due to its high chemical inertness, which is in accordance with the XPS results (Fig. S4). Therefore, the BN nanoflakes would perfectly keep in shape and isolate LPS solid electrolyte and lithium metal, suppressing the reaction and improving the interfacial stability.

The morphological structure of the LPS and LPS-BN solid electrolytes are tested and the TEM images are exhibited in Fig. 3. The TEM image of LPS solid electrolyte shows that the LPS particles are around several hundred nanometers with uniform size distribution (Fig. 3a). After BN doping, the LPS solid electrolyte would be coated with BN flakes, as exhibited in Fig. S1b-d. And with increase of BN content, the LPS particles aggregated together with wrapping a thicker and thicker BN flake. With BN nanoflakes coating, the reaction of LPS solid electrolyte and lithium metal could

be effectively suppressed in all-solid-state Li/S battery, thus increasing the interface compatibility of the batteries.

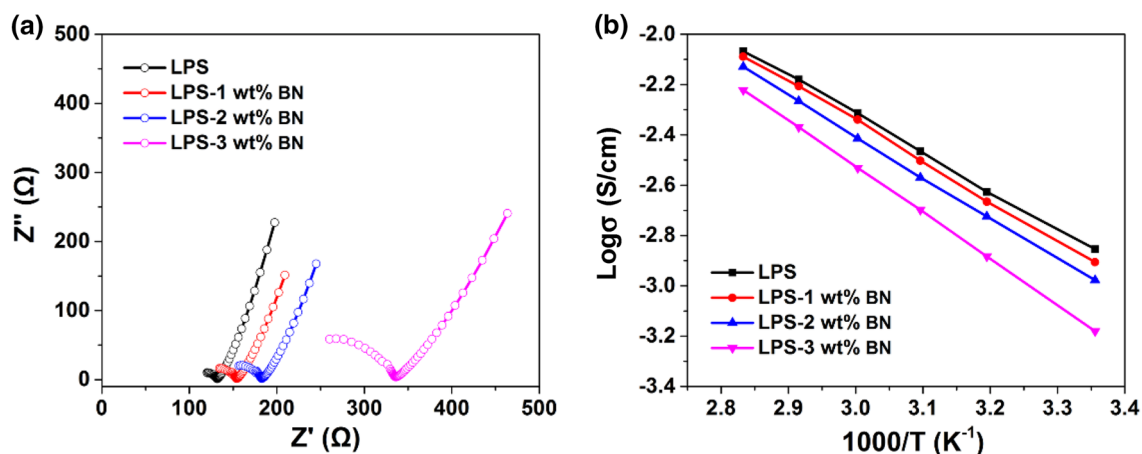
The ionic conductivity for the LPS and LPS-BN solid electrolytes is obtained by electrochemical impedance spectra (EIS) measurement method and the corresponding spectra is shown in Fig. 4a. For the ionic conductivity measurement, the symmetrical cell of  $\text{In}/\text{Li}_7\text{P}_3\text{S}_{11}/\text{In}$  cells were fabricated with indium (In) foil used as the electrode. As displayed in Fig. 4a, the ionic conductivity of the as-prepared LPS solid electrolyte is 1.4 mS/cm at room temperature. After BN doping, the ionic conductivity of the sulfide solid electrolyte decreases. And it becomes lower with increase of BN contents, which probably results from the fact that BN is an insulator of the electronic and ionic. The nano BN flakes could impend lithium-ion transfer after doping in LPS sulfide solid electrolyte, increasing the interfacial resistance between LPS particles and BN flakes. Therefore, the ionic conductivity of the solid electrolytes decreases after BN doped and it shows lower with higher nano BN flakes content.

The ionic conductivity of LPS and LPS-BN solid electrolytes have been measured in the temperature range from 298 to 353 K. Figure 4b shows the temperature dependence of ionic conductivity for LPS and LPS-BN solid electrolytes. As illustrated in the spectra, the dependences of conductivity on temperature obey the Arrhenius equation ( $\sigma = A \exp(-E_a/kT)$ ) for all the sulfide solid electrolytes. Moreover, the activation energies ( $E_a$ ) for lithium-ion conduction could be calculated from the slope of the lines. The calculated activation energy of as-prepared LPS solid electrolyte in the corresponding temperature range is 27.25 kJ/mol while the  $E_a$  for the LPS- $x$  wt% BN ( $x = 1, 2, 3$ ) solid electrolytes is 28.32, 29.97 and 31.05 kJ/mol, respectively. The results further prove that BN nanoflakes would deduce the ionic conductivity of LPS sulfide solid electrolyte as it could enlarge

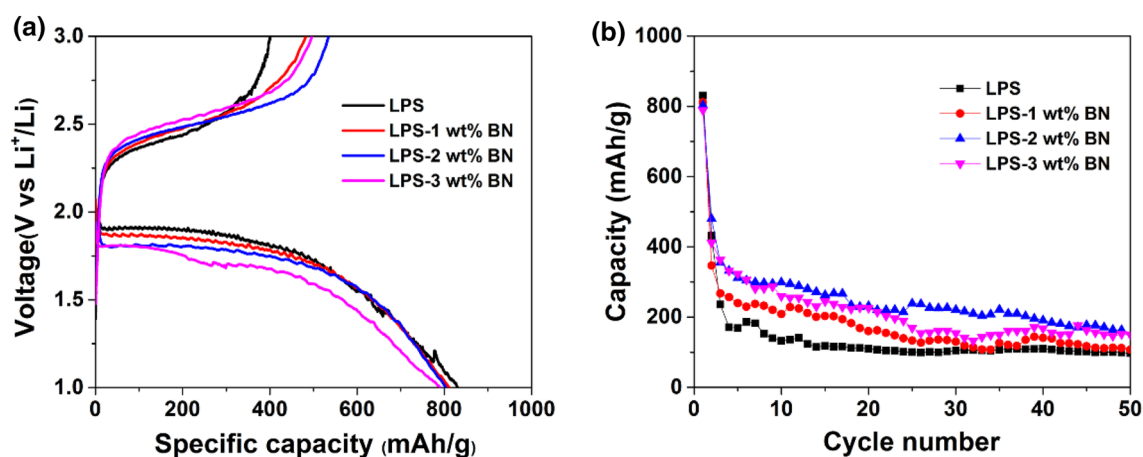
the interfacial resistance and impend lithium-ion diffusion and transmission.

To demonstrate the effect of BN doping on the LPS solid electrolyte interfacial stability with lithium metal, all-solid-state Li/S batteries with LPS and LPS-BN electrolytes were fabricated. The S, LPS and carbon black composites were ball milling and used as cathode. After the cathode composites and LPS solid electrolyte were loaded into the die sequentially and cold pressed at a certain pressure, the lithium foil was attached on the other side of the solid electrolyte surface and used as anode. The electrochemical performance of the all-solid-state Li/S batteries based on LPS and LPS-BN solid electrolytes were measured and compared, as presented in Fig. 5. The Fig. 5a exhibits the galvanostatic charge–discharge profiles of all-solid-state Li/S batteries with different solid electrolytes at a rate of  $C/20$  in voltage window of 1.0–3.0 V. Different from the conventional liquid Li/S cells, the curves for the all-solid-state Li/S batteries show only one platform of charge and discharge, which is in accordance with the CV results of all-solid-state Li/S batteries (Fig. S5). It is ascribed to the different reaction mechanism for the Li/S batteries with liquid and solid electrolyte. As we all know, there are two typical charge and discharge plateau in the liquid Li/S battery [9, 20, 21]. For the all-solid-state Li/S battery, no polysulfide intermediates formed during electrochemical cycle with use of solid electrolyte. Therefore, the all-solid-state Li/S battery only had an inconspicuous charge and discharge plateau corresponding to the reversible electrochemical reaction:  $\text{S} + 2\text{Li}^+ + 2\text{e}^- \leftrightarrow \text{Li}_2\text{S}$  [9, 20].

The initial discharge capacity of all-solid-state Li/S battery with LPS is 831.2 mAh/g, while 811.2, 803.5, 789.6 mAh/g for the cells based on LPS-1% BN, LPS-2% BN and LPS-3% BN solid electrolytes, respectively. It reveals that the ionic conductivity of the electrolyte would influence the



**Fig. 4** **a** Impedance plots of LPS and LPS- $x$  wt% BN with  $x$  varies from 1 to 3% at room temperature. **b** temperature dependent ionic conductivities of LPS and LPS- $x$  wt% BN with  $x$  varies from 1 to 3%



**Fig. 5** **a** The charge/discharge voltage profiles of all solid-state Li/S batteries based on LPS and LPS-*x* wt% BN with *x* varies from 1 to 3% at a rate of C/20 at room temperature. **b** Cycling performances

electrochemical of the batteries, as low ionic conductivity could impend lithium-ion transfer and decrease the capacity. However, the initial charge capacity of all-solid-state Li/S batteries deliver 400.6, 482.5, 646.5, 495.2 mAh/g for the Li/S cells with LPS, LPS-1% BN, LPS-2% BN and LPS-3% BN solid electrolytes. And the initial Coulomb efficiency is 48.1%, 59.4%, 80.4% and 62.7%, respectively. The low initial Coulomb efficiency for the Li/S battery with LPS solid electrolyte is probably caused by the reaction of LPS and lithium anode. BN nanoflakes are coated on the LPS particles and could effectively isolate the direct contact between LPS and lithium metal, thus suppressing the reaction of solid electrolyte and anode. Theoretically speaking, with more BN doping, the LPS could be wrapped more completely and the interface reaction would be suppressed more effectively, which is beneficial for the electrochemical performance of the all-solid-state Li/S batteries. However, as discussed in the previous section, the ionic conductivity of the solid electrolyte decreases with increase of BN content, which has a negative effect on the electrochemical performance of Li/S cells. Therefore, considering the ionic conductivity and interface compatibility of the solid electrolyte, the LPS with 2% BN doping would be the optimal for the all-solid-state Li/S batteries with the highest initial Coulomb efficiency and best cycling stability, as shown in Fig. 5.

Figure 5b show the electrochemical performance of the all-solid-state Li/S batteries with LPS and LPS-BN electrolytes. After 50 cycles, the discharge capacity of battery with LPS is 91.5 mAh/g, while 102.3, 185.4, 147.6 mAh/g for that based on LPS-*x* wt% BN electrolytes (*x* = 1, 2 and 3) respectively. It clearly indicates that the LPS-BN electrolytes would successfully improve the cycling stability of the Li/S cells, as the BN nanoflakes coating could inhibit the interfacial reaction. However, the ionic conductivity reduces with BN content increase,

of the all-solid-state Li/S batteries based on LPS and LPS-*x* wt% BN electrolytes (*x* varies from 1 to 3%) at a rate of C/20 at room temperature

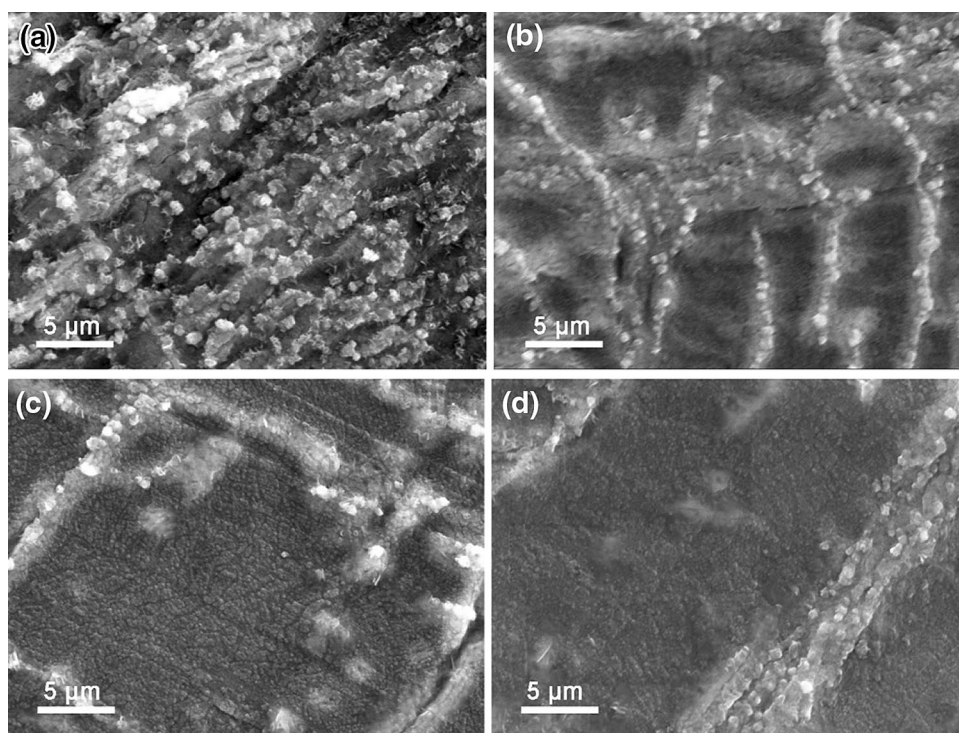
adverse to the cell electrochemical performance. Herein, the cycling stability of the Li/S increase with increase of BN from 1 to 2% and decrease with 3% BN content. The enhanced electrochemical performance of cell with LPS-BN electrolyte is attributed to its better interfacial compatibility with lithium anode due to BN nanoflakes coating and its relatively high ionic conductivity.

To further explore the effect of LPS-BN on interfacial compatibility with lithium anode, the morphology of lithium metal after 50 cycles in all-solid-state Li/S cells with LPS and LPS-BN solid electrolytes have been investigated, as shown in Fig. 6. From the SEM images, it clearly implies that the Li foil shows rough surface after 50 cycles of all-solid-state Li/S battery with LPS solid electrolyte, due to the reaction of LPS solid electrolyte and Li anode during cycling. However, the Li anode surface becomes smoother after 50 cycles with increase of BN content for the solid electrolyte in Li/S cells. It results from the fact that the LPS particles could not be coated perfectly with a low content of BN nanoflakes. With increase of BN doping, the BN would wrap the LPS particles successfully and effectively isolate Li mental from LPS solid electrolyte and prevent undesirable chemical reaction at the interface. Therefore, the surface of the Li anode is much smoother after 50 cycles in the Li/S cells based on LPS-BN solid electrolyte. It also further reveals that LPS-BN would increase the interface compatibility in all-solid-state Li/S batteries, thus increasing its electrochemical performance.

## 4 Conclusions

In summary, the BN nanoflakes doped LPS solid electrolytes with a various content were synthesized, forming a BN coating LPS structure. As an electronic and ionic insulator,

**Fig. 6** SEM images of Li foil after 50 cycles in the all-solid-state batteries based on LPS and LPS-*x* wt% BN solid electrolytes with *x* is **a** 0%, **b** 1%, **c** 2%, **d** 3%



BN doping would decrease the ionic conductivity of the solid electrolyte, which has a negative effect on the electrochemical performance of the Li/S batteries. However, with wrapped with BN nanoflakes, LPS would effectively isolate the direct contact with lithium metal in all-solid-state Li/S batteries, suppressing the reaction of LPS and lithium anode. Therefore, with improved interfacial compatibility of solid electrolyte and lithium anode, the all-solid-state Li/S battery with LPS-BN would exhibit a higher Coulomb efficiency and enhanced cycling stability at room temperature. Furthermore, combined with the ionic conductivity and interface stability caused by BN nanoflakes doping, it was found that the all-solid-state Li/S battery based on LPS-2% BN solid electrolyte delivered the optimal performance.

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